VisionGuard: Technical Review Script

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Length: about 6–7 minutes. Matches the actual wiring (Trig A0, Echo A1, moisture A2, buzzer D13) and the new firmware.

Section	Speaker	Time
1. System overview	Darshan	40 s
2. Ultrasonic ranging	Darshan	60 s
3. Water sensing	Darshan	50 s
4. Processing core	Tanay	50 s
5. Firmware architecture	Tanay	70 s
6. Power subsystem	Tanay	50 s
7. Hardware integration	Darshan	40 s
8. Live demo	Darshan	40 s
9. Limitations and future work	Tanay	40 s

1. System overview

Darshan · 40 s

> "Good morning, sir. I'm S Darshan, 25BCE2883, and this is Tanay Agarwal, 25BCE2475. Our project, **VisionGuard**, is an embedded electronic travel aid built on an **ATmega328P** microcontroller.
>
> It's a classic **sense–process–actuate** system. There are two sensing channels: **ultrasonic time-of-flight ranging** for obstacles and **resistive conductivity sensing** for water at the tip. A **priority-based state machine** drives an active buzzer. Objective: obstacle warning within **50 cm**, with distance encoded in the **beep repetition rate**, and a **continuous alarm for water that pre-empts obstacle alerts**. Hardware cost: about **₹876**."

2. Ultrasonic ranging subsystem

Darshan · 60 s

> "The **HC-SR04** is interfaced on **A0 as Trig** and **A1 as Echo**, using the analog pins as digital I/O (D14 and D15).
>
> The firmware asserts Trig HIGH for **10 µs**. The module transmits an **eight-cycle 40 kHz burst** and holds Echo HIGH for the **round-trip time of flight**. We measure the pulse width with `pulseIn()` and a **30 ms timeout**. The maximum range of 400 cm corresponds to about **23.5 ms** round trip; without the timeout, `pulseIn()` would block for 1 second.
>
> Distance is **d = v·t ÷ 2**, with **v ≈ 343 m/s = 0.034 cm/µs**, halved for the round trip. Theoretical resolution is **0.17 mm per µs**; we quantise to centimetres. We ping every **60 ms**, as the datasheet recommends, to avoid residual echoes, and apply a **3-sample median filter**, which rejects single-sample outliers like missed echoes."

3. Water-sensing subsystem

Darshan · 50 s

> "The probe's two exposed copper tracks form a variable resistance **R_p**. Inside the **LM393 module** it makes a **voltage divider** with a fixed resistor: **$V_{out} = V_{cc} \cdot R_p \div (R_p + R_f)$**. We read the module's analog output on **A2** with the **10-bit successive-approximation ADC**: 1024 levels, about **4.9 mV per LSB** at 5 V, about **104 µs per conversion**.
>
> Because the divider and the ADC reference share **AVcc**, the measurement is **ratiometric**: it doesn't change with battery voltage. Dry reads about **1023**; conductive water lowers it. We average **4 samples** and apply **hysteresis**: assert at **500**, release at **550**, which eliminates chatter near the threshold. The comparator's digital output, D0, is unused because the threshold is set in software."

Handover: "Tanay will explain the processing and firmware."

4. Processing core

Tanay · 50 s

> "The **ATmega328P** is an **8-bit AVR RISC** core with a **Harvard architecture**: separate buses for flash and SRAM, allowing a **two-stage fetch-execute pipeline**. It has **32 general-purpose registers**, executes most instructions in **one cycle**, and reaches about **16 MIPS at 16 MHz**.
>
> Memory: **32 KB flash**, about 2 KB of it reserved for the **bootloader**; **2 KB SRAM**; **1 KB EEPROM**. An earlier build of our firmware used about **3.7 KB of flash (12%)** and **281 bytes of SRAM (13%)**, leaving plenty of headroom. Code is compiled with **avr-gcc**, flashed with **avrdude** through the **CH340 USB–UART bridge** to the bootloader at **115200 baud**, and the application uses **9600 baud** for telemetry."

5. Firmware architecture

Tanay · 70 s

"The firmware is a **bare-metal, non-blocking super-loop**, a **soft real-time** design. Instead of `delay()`, every task is **time-sliced with `millis()`**, which is driven by the **Timer0 overflow interrupt** about every 1.024 ms. Unsigned subtraction keeps the timing correct across the **49.7-day rollover**.

Each pass of the loop: ranging runs every 60 ms, moisture is sampled every pass, the supply is checked every 10 s, and telemetry is printed every 200 ms. A **three-state FSM**, WATER, OBSTACLE and CLEAR, is evaluated with **WATER checked first**, which enforces priority.

In OBSTACLE, the beep gap is a **linear map**: **gap = (d − 2)·350 ÷ 48 + 50 ms**, so 2 cm gives 50 ms and 50 cm gives 400 ms. Because nothing blocks, the **worst-case water-alarm latency dropped from about 510 ms to about 30 ms**, which is bounded by one `pulseIn()` call.

At start-up, a **self-test** pings five times, reports the sensors and supply, and plays a ready chime. In CLEAR, a **30-second heartbeat chirp** confirms to a blind user that the cane is powered."

6. Power subsystem

Tanay · 50 s

"Power comes from a single **IMR 18650 Li-ion cell**: **3.7 V nominal, 1200 mAh, 4.44 Wh**. It goes through a rocker switch **directly to the 5V rail**, bypassing the **AMS1117 regulator**, because VIN needs about 7 V. A **220 µF bulk capacitor** absorbs transient current when the buzzer switches.

One trade-off: the datasheet guarantees **16 MHz only from about 3.8 V**, so a drained cell is marginal, and the **brown-out detector** resets at about 2.7 V. To protect the cell, the firmware measures its own supply **without extra hardware**: it selects the **internal 1.1 V bandgap** as the ADC input, with AVcc as the reference, so **Vcc = 1.1 × 1023 ÷ reading**. Below **3.4 V** it plays a three-chirp low-battery warning, with hysteresis at 3.6 V. At about 40 mA, theoretical runtime is about **30 hours**."

7. Hardware integration

Darshan · 40 s

"We soldered **male headers** onto the Nano. Signal lines use **female jumpers** so modules can be serviced. The **power lines, the buzzer and the capacitor are soldered** for mechanical reliability: a cane is subject to constant vibration, so we didn't use a breadboard. The ultrasonic sensor takes 5 V and GND from the **ICSP header**. The **active buzzer is on D13**, sharing the on-board LED, with its return on the **USB shield ground**. D3, a PWM pin, is reserved for haptics."

8. Live demo

Darshan · 40 s

"Power on: you hear the **self-test chime**. CLEAR state: silent. As I bring my hand in, the **repetition rate increases** as distance falls. Now the tip in water: the **WATER state pre-empts** and the tone becomes continuous, even with my hand in range. The Serial Monitor shows distance, ADC value, supply voltage and state in real time."

9. Limitations and future work

Tanay · 40 s

> "Limitations: the sensor's **15° beam** misses head-level and thin obstacles; water is detected only on **contact**; DC excitation causes **electrolytic corrosion** of the probe; and the supply is **unregulated**.
>
> Next: **haptic feedback** on D3 through the **IRLZ44N logic-level MOSFET** with a **flyback diode**; a **boost converter, TP4056 charger and DW01 protection**; **Timer1 input capture** to replace the blocking `pulseIn()`; a **capacitive probe**; and angled sensors for head-level and drop-off detection. Thank you, sir. We're happy to take questions."

Numbers to have ready

Pins	Trig A0, Echo A1, moisture A2, buzzer D13, haptics D3
Ultrasonic	40 kHz, 10 µs trigger, 30 ms timeout, 60 ms ping, 2–400 cm, 15°
Formula	d = t × 0.034 ÷ 2
ADC	10-bit SAR, 4.9 mV/LSB, 104 µs
Water	On < 500, off > 550, 4-sample average
Battery	Vcc = 1125300 ÷ reading (mV), warn < 3400 mV
MCU	16 MHz; 32 KB flash / 2 KB SRAM / 1 KB EEPROM; Harvard RISC
Latency	510 ms → about 30 ms

Two honest points. (1) Say "**our earlier build used 3.7 KB**". The new code is larger, so check the exact size the Arduino IDE shows after you click ✓, and use that number. (2) Present the water demo only after you've **tested it with the new code**. If calibration isn't done yet, say: "The water threshold is being calibrated from Serial Monitor readings."